Last time, we reviewed some of the fundamentals of probability:

probability (what it is)
random variables
distributions of a random variable
$\mathbb{E}(X)$, $\text{var}(X)$ expected value and variance of a random variable

We also talked about relationships between random variables.

- joint dists $p(x, y)$
- conditional prob $\mathbb{P}(X = x \mid Y = y)$
- $\text{cov}(X, Y)$, $\text{correl}(x, y)$
- independence $X \perp\!\!\!\perp Y$

We also talked about some useful rules for expected value. Some of those rules concerned adding or multiplying by a constant.

$$\mathbb{E}(aX) = a\mathbb{E}(X) \qquad \mathbb{E}(X + b) = \mathbb{E}(X) + b$$

Another rule was that when we add two variables together, their expected values add.

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y) \text{ always}$$

Next, we'll review the corresponding rules for variances.

variance rules

When we multiply by a constant, the variance gets multiplied by that constant, squared.

$$\text{var}(aX) = a^2 \text{var}(X)$$

If we multiply X by a negative constant, the variance gets multiplied by a positive value, because we square it, and the variance will still be positive.

Adding a constant to a RV shifts the distribution to the left or right, but doesn't change how spread out it is, so the variance doesn't change.

$$\text{var}(X+b) = \text{var}(X)$$

On a related note, the variance of a constant simply equals zero, because the constant doesn't "vary":

$$\text{var}(b) = 0$$

When we add two random variables together, the variance of their sum depends on the association between the variables. Suppose X is a student's score on the math section of the SAT, and Y is their score on the reading section.

$\text{var}(X) = 100^2$
$\text{var}(Y) = 100^2$
} IF X, Y indep, variances add

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) = 100^2 + 100^2$$

If a student's scores on the 2 sections are independent, the variances add (the standard deviations never add).

But if the section scores are correlated instead of independent, we CAN'T just add them. In that case, the variance of their sum also depends on their correlation $\rho_{xy}$

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\rho_{xy}\sigma_x\sigma_y$$

(correl, sd(x), sd(y))

what does this mean? If scores on the 2 sections are positively correlated, then total scores are more spread out.

↳ someone above avg on math is also likely to be above average on reading, pushing their total score further away from the middle

If scores on the 2 sections are negatively correlated, total scores are less spread out.

↳ someone above avg on math is likely to be below avg on reading, so these offset and push their total score back toward the middle.

---

Today we're going to talk about some examples of common, well known types of probability distributions, starting with Bernoulli.

Bernoulli: $X \in \{0,1\}$

$\mathbb{P}(X=1) = p \quad \mathbb{P}(X=0) = 1-p$

A Bernoulli RV is a binary variable that takes the value 1 (the "success" outcome) with some fixed probability $p$. An example of a Bernoulli could be the outcome of a coin flip, where $p$ is the probability of heads. For a fair coin, $p = \frac{1}{2}$.

There is a whole "family" of Bernoulli distributions, each one indexed by the value of $p$, the success probability.

family of dists: Bern(0.1), Bern(0.2), etc.

For a Bernoulli with success probability $p$, what are the properties of that distribution?

$\mathbb{E}(X) = 0 * \mathbb{P}(X=0) + 1 * \mathbb{P}(X=1) = 0 + p = p$

The expected value is the probability-weighted average of the two possible outcomes: 0 and 1. We can calculate the variance similarly.

$$\sigma_X^2 = \text{var}(X) = \mathbb{P}(X=0)(0 - \mathbb{E}(X))^2 + \mathbb{P}(X=1)(1 - \mathbb{E}(X))^2$$
$$= (1-p)\,p^2 + p\,(1-p)^2$$
$$= p(1-p)$$

(we skipped some arithmetic here)

$$\text{sd}(X) = \sqrt{p(1-p)}$$

The Bernoulli distribution is a building block for the next example we'll discuss: the Binomial distribution.

The binomial distribution arises when we repeat multiple instances of the same Bernoulli trial, that are indep. of each other and all have the same probability of success. For example, we could flip a coin 5 times and observe each outcome.

$$X_1, X_2, \ldots, X_5 \overset{iid}{\sim} \text{Bern}\left(\tfrac{1}{2}\right)$$

iid = "indep, identically distributed"

Then the total number of heads in the collection of 5 coin flips is an example of a binomial RV.

define $Y = \sum_{i=1}^{5} X_i$ $\qquad Y \sim \text{Binomial}\left(n=5, p=\tfrac{1}{2}\right)$

The binomial distribution, just like the Bernoulli, is a family of distributions. You specify which dist. in that family you're talking about by giving a specific value of n, the number of trials, and p, the prob. of success in each trial.

Binom $(n, p)$

Now let's consider the properties of the binomial distribution. To get the expected value and variance, we can use the rules about sums of independent random variables.

$$\mathbb{E}(Y) = \mathbb{E}(X_1 + \dots + X_5) = \mathbb{E}(X_1) + \dots + \mathbb{E}(X_5)$$

The total # of heads, Y, is the sum of all the {0, 1} outcomes of the 5 individual coin flips, each with expected value equal to p.

$$= \underbrace{p + \dots + p}_{5 \text{ times}} = 5p$$

We're using $p = \frac{1}{2}$ for a fair coin, so ...

$$= 5\left(\tfrac{1}{2}\right) = 2.5$$

if we flip a fair coin 5 times, the expected # of heads is 2.5.

For arbitrary values of n and p, $\mathbb{E}(Y) = n * p$.

$$\mathbb{E}(Y) = np$$

we can find the variance in the same way.

$$\begin{aligned} \text{var}(Y) &= \text{var}(X_1 + \dots + X_5) \\ &= \text{var}(X_1) + \dots + \text{var}(X_5) \quad \text{indep} \\ &= 5 * p * (1-p) \end{aligned}$$

For a fair coin, the variance of # heads in 5 flips is:

$$5\left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right) = 1.25$$

and more generally, for n trials w/ success probability p:

$$\text{var}(Y) = np(1-p)$$

The expected value and variance summarize the distribution of Y, but what is the actual probability distribution? What is the probability of each possible # of heads we could get?

| $y$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(Y=y)$ | | | | | | |

Finding these probabilities is a counting problem - it requires us to think about all possible sequences of 5 coin flips, and enumerate how many of them have the desired number of heads.

$P(1,1,1,0,0)$

Consider the specific outcome where we get 3 heads followed by 2 tails.

The probability of that specific outcome is

$$\underbrace{p}_{\text{heads 1st time}} * \underbrace{p}_{\text{heads 2nd time}} * \underbrace{p}_{\text{heads 3rd time}} * \underbrace{(1-p)}_{\text{tails 4th}} * \underbrace{(1-p)}_{\text{tails 5th}}$$

each flip is independent, so we multiply the probabilities

$$p^3(1-p)^2 = \left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = \frac{1}{32}$$

That's <u>one</u> of the possible ways to get 3 heads in 5 flips. Any sequence with 3H and 2T has this same probability - rearranging the order doesn't change that.

So the total prob. of exactly 3 heads is the # of sequences with 3 heads, times the probability of each individual sequence

$$P(Y=3) = \left(\begin{array}{c}\#\text{ sequences} \\ \text{with 3H}\end{array}\right) * p^3(1-p)^2$$

The number of ways to get exactly 3 heads in 5 flips is denoted 5 choose 3 $\binom{5}{3}$

$$\mathbb{P}(Y=3) = \binom{5}{3} p^3 (1-p)^2 \qquad \binom{5}{3} = \frac{5!}{3!(5-3)!}$$

more generally:

$X \sim \text{Binom}(n, p)$

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

~GO TO NOTEBOOK~

Suppose 15% of adults are left handed.
We randomly select 10 people, and ask whether they're left-handed.
For each person, $P(X=1) = 0.15$ the probability is

$X = \begin{cases} 1 & \text{LH} \\ 0 & \text{RH} \end{cases}$ $\quad P(X=1) = 0.15$

Survey 10 people $X_1, \dots, X_{10}$

$Y = \sum_{i=1}^{10} X_i$ Then the number of left handed people in a random sample of 10 follows a binomial distribution, with $n=10$ and prob $p=0.15$

$Y \sim \text{Binom}(10, 0.15)$

What is the probability that exactly one person in the sample is left handed?

$$\mathbb{P}(Y=1) = \binom{10}{1} 0.15^1 * (1-0.15)^{10-1}$$ dbinom(1,10,0.15)

What is the probability that at most 2 people in the sample are left handed?

$$\mathbb{P}(Y \le 2) = \mathbb{P}(Y=0) + \mathbb{P}(Y=1) + \mathbb{P}(Y=2)$$
$$= \binom{10}{0} 0.15^0 0.85^{10} + \binom{10}{1} 0.15^1 0.85^9 + \binom{10}{2} 0.15^2 0.85^9$$

= dbinom(0,10,0.15) + dbinom(1,10,0.15) + dbinom(2,10,0.15)
= pbinom(2,10,0.15) ← cumulative distribution fn

What is the 95th percentile of the distribution?

Find the smallest number k such that the probability of having at most k left-handed people is $\geq 95\%$.

$P(Y \leq k) \geq 0.95, \quad k = ?$

qbinom(0.95, 10, 0.15)

R also has built-in tools for drawing random samples from a binomial distribution. We'll look at how to do that, and also use this as an opportunity to talk about the difference between a probability dist. and sample data.

expected value $E(X)$ and variance $var(X), \sigma_x^2$ are properties of a probability distribution, which is theoretical.

If we observe a set of values sampled from that distribution, the sampled values have their own sample mean and sample variance.

sample mean $\bar{x}_n = \frac{1}{n} \sum_{i=1}^{n} x_i$

sample variance $s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x}_n)^2$

sample sd $= \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x}_n)^2}$

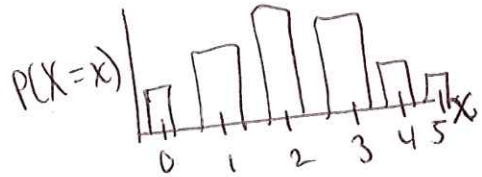
Imagine we flip a pair coin 5 times and record the # of heads. Let that number be $Y_1$. Then, repeat that experiment 9 more times, recording the # of heads each time.

Now we have sample values $Y_1, Y_2, \ldots, Y_{10}$
drawn from a binomial dist with $n=5$ and $p=0.5$.
The expected value of the distribution is n times p,
$5 * \frac{1}{2} = 2.5$., and variance $n*p*(1-p) = 5(\frac{1}{2})(\frac{1}{2}) = 1.25$.
The sample we observed has its own mean & variance,
which we can calculate.

We could repeat this whole exercise & sample 10 new values,
and those values would have a new mean & variance.

The expected value & variance of the underlying distribution
stay the same.

We can also compare the sampled data to the underlying
binomial distribution visually.
We can visualize the binomial dist with a bargraph
displaying the probability of each possible outcome

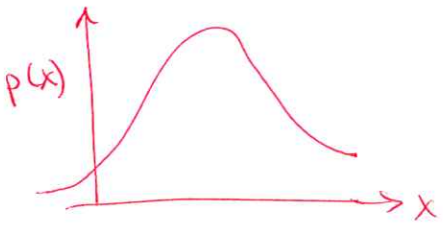



For the sampled data, we can create a similar
visualization by making a histogram. Instead of
probability on the y-axis, the height of the bars
represent how frequently that value occurred in the
observed sample.

~back to the board~

Next we're going to move on to a different distribution, the normal distribution (also called "Gaussian").

Normal (gaussian)
* continuous
* symmetric
* bell curve




this is a distribution for continuous random variables, and its probability density function has this familiar symmetric, shape that we say looks like a bell curve.

Like binomial, this is a family of distributions. A binomial distribution was specified by the number of trials $n$ & prob. of success $p$.

A normal distribution is specified by its expected value $\mu$ and its variance $\sigma^2$

family of dists
parameterized by exp. val $\mu$
variance $\sigma^2$
$\} N(\mu, \sigma^2)$

Because it's continuous, the probability of taking any exact value is zero - like we mentioned before, if

X is an adult's blood pressure and we say

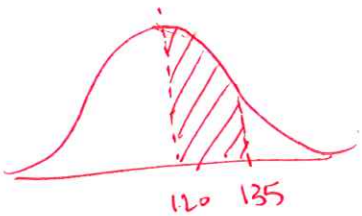
$X = BP$ $\quad X \sim N(120, 15^2)$
$\uparrow \sigma^2 = 15^2, \sigma = 15$

The probability of blood pressure being exactly 120 is zero

$\mathbb{P}(X=120)=0$, but we CAN talk about the probability of blood pressure being in a certain interval

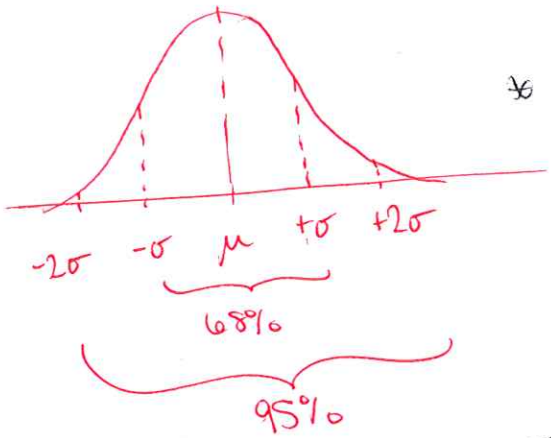
for example $\mathbb{P}(120 \le X < 135) = 0.34$

The probability of X taking a value in that interval corresponds to the area under the pdf curve.




This is something you look up in a table or on a computer, not a calculation you do by hand.

but there are a few special rules about area under the curve that you could memorize.




* prob. of being within 1 standard deviation of the expected value $\mu$ is 68%.
* (more useful) prob of being within 2 std dev is 95%.

---

Normal distributions have some special properties to be aware of. If X is normal, then linear transformations of X are also normal.

properties ① $X \sim N(\mu, \sigma^2) \Rightarrow \{aX+b\} \sim N(a\mu+b,\ a^2\sigma^2)$

For other types of probability dists, this is generally NOT true. Now, that might be confusing because we mentioned that expected value is linear.

$E(aZ+b) = aE(Z)+b$ for arbitrary $Z$

and that's true for any random variable.

But it's not necessarily true that the linear transformation belongs to the same family of probability distributions.

(aside: one obvious counterexample is the binomial dist. if $Y \sim Binom(n, p)$ and we define a new variable $W \sim 2*Y$, then $W$ can only take even-number values. A binomial distribution takes values $\{0, 1, 2, \ldots, n\}$, so $W$ can't be binomial.)

② if $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$ & $X \perp\!\!\!\perp Y$

$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$

Another special property of normal distributions is that the sum of independent normal RVs is also normal.

Now let's look at some computational things we can do with the normal distribution in R. ~ GO TO NOTEBOOK ~

R has built in function for the probability density function, which we can graph

`dnorm(x, mean, sd)`

It also has a cumulative distribution function,

For evaluating the prob. that the RV is less than or equal to a certain value.

$\mathbb{P}(X \leq a)$ pnorm(a, mean, sd)

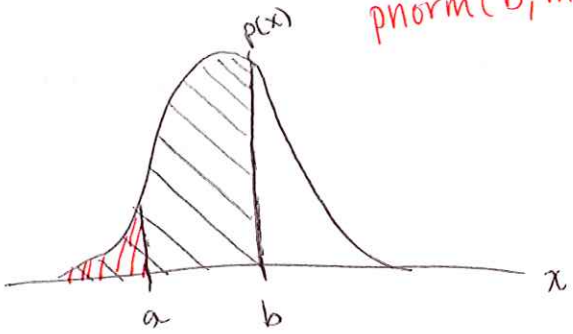
For the probability of being GREATER than a certain value, subtract the cumulative probability from 1:

$\mathbb{P}(X > a) =$ 1 - pnorm(a, mean, sd)

To calculate the probability of being in the interval [a, b], take the probability of being less than [b], and subtract the probability of being less than [a]

$\mathbb{P}(a < X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a)$

pnorm(b, mean, sd) - pnorm(a, mean, sd)




We can also get quantiles. What is the 95th percentile - that is, smallest value of b such that $\mathbb{P}(X \leq b) \geq 0.95$?

↳ for binomial you can get this by calculating the cumulative probability of each value of x

↳ for a continuous variable, you can't do that
↳ use the quantile function qnorm(0.95, mean, sd)

R also has a built in function for generating random samples from a normal distribution

```
rnorm(n, mean, sd)
      ↑
# samples to generate
```